

Unit - I HyperText Markup Language (HTML)

Introduction to HTML - HTML Document structure -
Text - Lists - Frames, Tables and Div tags -
Forms - Images - Hyperlinks - Tag Attributes -
Structuring webpage - Semantic HTML.

Introduction to html:

* HTML is a markup language which provides a set of tags suitable for making up webpages.

* HTML is a scripting language for web Pages. and output of programs can be seen after using browser.

* HTML is a hypertext markup language because it supports font style text, pictures, graphics and animations.

* The HTML documents are plain text files. These are created by using text Editors like notepad.

* HTML is a tagbased system. A tag is a special instruction for browser.

* A tag is made up of left operator (<) and right operator (>) and a tagname between these two operators optionally tagname contains one or more parameters.

* The browser won't generate any errors.

Syntax: <tagname [parameter = Value]>

* The latest Version is HTML 5 which allows developers to embed a video or audio in the websites.

Note: Dont give any space between left operator and Tag name.

* In HTML every Tag can have corresponding ending Tag preceded by forward slash Symbol.

ADVANTAGES	DISADVANTAGES
i). HTML is used to build website.	i). HTML can only create static webpages.
ii). It is supported by all browsers.	ii). A large amount of code has to be written to create a simple webpage.
iii). It can be Integrated with other languages like CSS, Javascript etc....	iii). Editing of webpage need to be done seperately.

HTML Document Structure:

* All HTML Document, does follow some basic structure.

* It has two Blocks

→ Head Block

→ Body Block.

* Head block contains control information used by the browser and title of the document.

<html>... </html>

* This tag is the root element of HTML document. By using this tag the browser can identify whether it is html document or not.

<head>... </head>

* This tag indicates the first part of HTML document and it contains meta information and title of the html document.

<title>... </title>

* This tag specifies the title of html document which is shown in the browser's title bar or in the page tab.

<meta>... </meta>

* After <head> line <meta> element should be used.

* There can be multiple <meta> lines in the same webpage.

* It provides additional information such as page keyword, page description and authors of a web document.

Syntax:

<head>

<meta http-equiv="content-type"
Content="text/html; charset=utf-8"/>

<title> page title goes here </title>

</head>

http-equiv: It provides an http header for the information/Value of the content attribute.

Content-type: It specifies the character encoding for the HTML document.

Charset: It specifies the character encoding for the document.

utf-8: It specifies the webpage character encoding.

<body>... </body>

* It indicates the second part of HTML document and contains all the remaining information about the webpage. The body tag has different parameters which indicates background, bgcolor... etc

Blocks (<p>... </p>, heading tags)

* The two major blocks of text in HTML are paragraph and headings.

Paragraph Syntax: <p [align = "left" | "center" | "right"]>... </p>

Heading Syntax: <h1>... </h1> <h3>... </h3>
<h2>... </h2> <h4>... </h4>

HTML Document Structure Example

Program

```
<!DOCTYPE html>
<html>
<body>
<h1>Chennai Institute of Technology</h1>
<h2>Chennai Institute of Technology</h2>
<h3>Chennai Institute of Technology</h3>
<h4>Chennai Institute of Technology</h4>
<h5>Chennai Institute of Technology</h5>
<h6>Chennai Institute of Technology</h6>
</body>
</html>
```

Output

Chennai Institute of Technology

Chennai Institute of Technology

Chennai Institute of Technology

Chennai Institute of Technology

Chennai Institute of Technology

Chennai Institute of Technology

Text Element: HTML contains several elements for defining text with a special meaning.

HTML Formatting Elements:

i).

* By using this tag we can set the size and color of text.

ii) ... - It bolds the text.

iii) ... - for strong (new standard for bold).

iv). <u>...</u> - It underlines the text.

v). <i>...</i> - Italic text.

vi). ... - for emphasis (new standard for Italic).

vii). <s>...</s> - Strikes the text.

viii) _{...} - Subscript text.

ix). ^{...} - Superscript text.

Program

```
<!DOCTYPE html>
<html> <head>
<title>Text formatted document</title>
</head>
<body>
<p> Application Development
Practices</p><br>
<b>Static web page using HTML</b>
<br>
<strong> HTML-Hypertext Markup
Language</strong><br>
<i>This text is italic.</i><br>
<em> This text is emphasized </em><br>
<mark>Marked text </mark><br>
<small>Smaller text </small><br>
<del>Deleted text </del><br>
<ins>Inserted text</ins><br>
<p>HTML<sub>Subscript text </sub></p>
<p>HTML<sup>Superscript text
</sup></p><br>
</body>
</html>
```

Output

Text formatted document

Application Development Practices

Static web page using HTML

HTML - Hypertext Markup Language

This text is italic.

This text is emphasized

Marked text

Smaller text

~~Deleted text~~

Inserted text

HTML_{Subscript text}

HTML^{Superscript text}

unclosed HTML Tags'. Some HTML Tags are not closed, for example br, hr.

 Tag: br stands for break line, it breaks the line of the code.

<hr> Tag: hr stands for horizontal rule. It is used to put a line across the webpage.

Syntax: <hr [width=n] [size=n]

[align = {left/right/center}]

[Noshade]]/>

Horizontal rule Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<p> There is a horizontal rule below this  
Paragraph </p> <hr>
```

```
<p> This is horizontal rule! </p>
```

```
</body>
```

```
</html>
```

Output:

There is a horizontal rule below this
Paragraph

This is horizontal rule!

Character escape Sequences:

In HTML certain symbols are treated as special.

character	Meaning	Entity Reference
<	Less than	<
>	greater than	>
&	ampersand	&
//	double quote	"
'	apostrophes	'
©	copyright	©

Anchor tag <a>: Anchor tag is used to create link between two documents, minimum it requires a parameter ie) "href" which indicates destination document.

* The most important capability of html is its ability to create hyperlinks.

* Hyperlinks can be applied to either text or images.

Syntax: Text

Example:

<p> you can reach michael at </p>

website

Email

phone

Output:

You can reach michal at

- website
- Email
- phone

Lists: List is nothing but the collections of items or elements. There are two types of Lists, ordered List and unordered List. 3 types of list available in html.

- Syntax
- i). ordered List - `<ol>...</ol>`
 - ii). unordered List - `<ul>...</ul>`
 - iii). Definition List - `<dl>...</dl>`

i). ordered List: This is used to display the List of items in a order. It can use the numbers or roman numbers or alphabets as type of orders.

* It has one inbuild tag ie) `<li>...</li>`
In this list the default order is numeric.

Example1	Example2	Example3
<pre> orange grapes apple </pre> <u>O/P</u> 1. orange 2. grapes 3. apple.	<pre><ol type = 'A'> mango orange </pre> <u>O/P</u> A. mango B. orange	<pre><ol type = 'I start = 3'> cricket football </pre> <u>O/P</u> III. cricket IV. football

ii). unordered List: This is used to display the list of item by using different types of

Symbols. In this list also have a inbuilt tag ie) `<li> ... </li>` In this list default symbol is dot (•)

Example 1	Example 2
<pre> orange Apple </pre> <u>O/P</u> • orange • Apple	<pre> <ul type = "circle"> cricket Football </pre> <u>O/P</u> ○ cricket ○ football

ii) Definition list: The definition list is used to specify list of terms and their definition. It has the following tags.

`<dl>` tag specifies the definition list.

`<dt>` tag specifies the definition term.

`<dd>` tag specifies the description for term.

Example:

<pre> <dl> <dt> HTML </dt> <dd> HTML is markup language </dd> <dt> XML </dt> <dd> XML is extended Markup Language </dd> </dl> </pre>	<u>O/P</u> <pre> HTML HTML is a markup language XML XML is extended markup language </pre>
--	---

Program	Output
<pre> <!DOCTYPE html> <html> <body> <h2>An Unordered HTML List</h2> HTML CSS DOM Javascript <h2>An Ordered HTML List</h2> HTML CSS DOM Javascript <dt>Problem Solving</dt> <dl> <dd>- Using C Programming</dd> <dt>DBMS</dt> <dd>- Database Management Systems</dd> </dl> </body> </html> </pre>	An Unordered HTML List • HTML • CSS • DOM • Javascript An Ordered HTML List 1. HTML 2. CSS 3. DOM 4. Javascript Problem Solving - Using C Programming DBMS - Database Management Systems

Frames: Generally the browser window can be used to display the one or more documents at a time.

* The window can be divided into rectangular areas each of which is "frame".

<frameset> tag specifies the number of frames and their layout.

* Frameset tag takes the place of body tag ie) html document has either a body or a frameset tag, but can't have both.

* frameset tag must have either a "rows" or "cols" attribute and they often have both

Rows: No. of rows of frames that will occupy windows

Columns: No. of columns of frames that will occupy window.

Syntax:

```
<frameset [cols = "x.x." ] [rows = "x.x." ] >
```

```
</frameset>
```

The <frameset> tag have one inbuilt tag ie) <frame> tag. It is used to specify the content of frame.

```
<frame src = "filename" [name = ""] [scrolling =  
"yes/auto"] [frameborder  
= "0"/"1"] />
```

Frame Attributes:

src: It specifies source for frame.

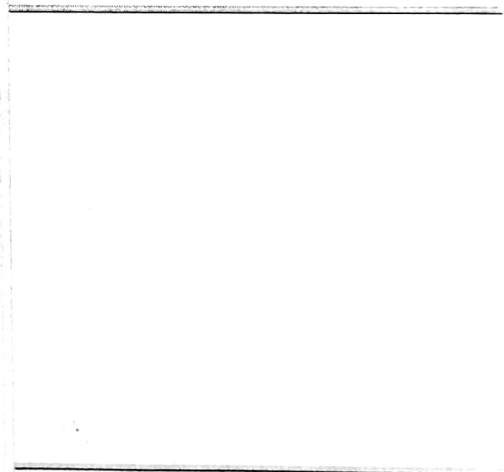
name: It specifies name of the frame.

Note: frame are not supported in HTML5
Instead iframe is used.

HTML Iframes Example Program

Output

```
<!DOCTYPE html>  
<html>  
<head>  
<title>Example of HTML Frames using row  
attribute</title>  
</head>  
<frameset rows = "20%, 60%, 20%"> <frame  
name = "top" src =  
"C:/Users/dharam/Desktop/attr1.png" />  
<frame name = "main" src =  
"C:/Users/dharam/Desktop/gradient3.png" />  
<frame name = "bottom" src =  
"C:/Users/dharam/Desktop/col_last.png" />  
</frameset>  
<body>The browser you are working does not  
support frames.</body>  
</frameset>  
</html>
```



Iframe: An HTML iframe is used to display a web page within a web page.

* The HTML <iframe> tag specifies an inline frame.

Syntax: `<iframe src="url" title="description">`

Height and width: use height and width attributes to specify the size of the iframe.

* Height and width is specified in pixels by default.

Syntax: `<iframe src="demo-iframe.html"`

Example: `height="200" width="300"`
`title="iframe Example">`
`</iframe>`

Tables: HTML table allow the developers to arrange data into rows and columns.

* The `<table>` tag in HTML is used to create tables on webpages.

* The tables are way to organize and display the data into rows and columns.

Basic Table Tags:

`<table>` → It defines a table.

`</table>`

`<tr>...</tr>` → It defines a row in table.

`<th>...</th>` → It defines a header cell in a table.

`<td>...</td>` → It defines a cell in table

`<caption>...</caption>` → It defines a table caption

`<colgroup>...</colgroup>` → It specifies a group of one or more columns in a table for formatting.

`<col>...</col>` It is used with `<colgroup>` element to specify the column properties for each column.

`<tbody>... </tbody>` → It is used to group the body content in a table.

`<thead>... </thead>` → It is used to group the header content in a table.

`<tfoot>... </tfoot>` → It is used to group the footer content in a table.

Syntax: `<table [align="center"/"left"]
[border=n] [cellpadding=n] [height=n%]
[width=n%] [cellspacing=n]>`

`<caption>... </caption>`

`<tr>`

`<th>... </th>`

`</tr>`

`<tr>`

`<td>... </td>`

`</tr>`

`</table>`

Cell padding: It specifies the space between content of the cell and its borders in pixels.
cell spacing: It specifies the space between cells.

width: It sets amount of screen that table will use in the factor of width.

Height: It sets amount of screen that table will use in the factor of height.

```
<!DOCTYPE html>
<html>
<head>
<style>
table, th, td {
border: 1px solid black;
}
</style>
</head>
<body>
<h1>The table element</h1>
<table>
<tr>
<th>Department</th>
<th>section</th>
</tr>
<tr>
<td>CSE</td>
<td>13</td>
</tr>
<tr>
<td>ECE</td>
<td>4</td>
</tr>
</table>
</body>
</html>
```

The table element

Department	section
CSE	13
ECE	4

Div Tags: Div tag is also known as division tag

- * HTML <div> tag is a non-semantic tag.
- * It is generally used by developer to group the HTML element together and apply css style to many element at once.
- * It helps to separate the data in web page and create particular section for a particular data or function in a webpage.
- * It also used in HTML to make division of content in the webpage (text, images, header, footer, navigation bar etc...).
- * <div> tag is generally used to group the content and provide css style using class or id attribute.

Example: `<div class="div1">`
`<h2>This is heading </h2>`
`<p>This is paragraph </p> </div>`

<div> Tag	<table> Tag
* It is more flexible * can modify in div Tag easily * Page rendering is high comparing to table	* It requires coding Skill. * Modification is difficult in table tag. * Page rendering is very slow.
<u>Forms:</u> <form> tag used to create an HTML form for user input. * Forms are used to pass user data to specified url. Attributes: * Method * Action. Method: This attribute specifies how the entered data in form is sent to destination. There are two methods of sending information one is "post" and other is "get". Action: This attribute specifies the url of the application which is to be called when the user clicks the submit button or submitting the form.	
"Post" Method	"GET" method
* Information sent along with body. * Data is invisible while sending. * It provides security * It can send any no. of characters	* Information sent along with URL * Data is visible in site * No security. * It sends limited no. of characters.

User Interface elements in forms:

- i). Label iv). Password Field vii). Combo Box
ii). Text field v). Radio Button viii). List box
iii). Buttons vi). Check Box ix). Text area

i). Label: It is used to display text on label screen.

ii). Text field: It is an rectangular box. It is used to take the input from the user. It is created using `<input>` tag.

Format `<input type="text" [size=n] [name=" "] [maxLength=n] [value="name"]>`
`</input>`

iii). Button: It supports three types of buttons
Button: `<input type="button" [value=" "]>`
`</input>`

Submit Button: `<input type="submit" [value=" "]>`
`</input>`

Reset Button: `<input type="Reset" [value=" "]>`
`</input>`

iv). Password field: It is used to enter the password.

`<input type="password" [size=n] [name=" "]>`
`</input>`

v). Radio Button: It is used to select only one value at a time.

`<input type="radio" [name=" "] [value=" "]>`
`</input>`

vi). check Box: It is used to select multiple value at a time.

`<input type="checkbox" [name=" "] [value=" "]>`
`</input>`

vii). Combo Box: It is used to select one value from.

```
<select [name=" "]>
```

```
<option> . . . </option>
```

```
<option> . . . </option>
```

```
</select>
```

viii). List Box: It is used to select one/more Value from multiple Value.

```
<select [name=" "] [size=n]>
```

```
<option [selected]> . . . </option>
```

```
<option> . . . </option>
```

```
</select>
```

ix). Text area: It is text like text field it has multiple rows and columns.

```
<textarea [name=" "] [rows=n] [cols=n]>
```

```
</textarea>
```

FORMS EXAMPLE

Program

```
<!DOCTYPE html>
<html lang="en">
<head> <title>Forms</title> </head>
<div>
<div align="center">College Registration</div>
<form> <fieldset>
<legend>Personal Details</legend>
<label><B>Name: </B><input name="Name"></label> <br /><br />
<label><B> Gender: M </B><input type="radio" name="Male"> F <input type="radio"
name="Female"></label> <br /><br />
<label><B> Address: </B></label>
<textarea id="Address" name="Address" rows="5" cols="35"></textarea>
<br /><br />
<label><B> DOB: </B><input type="date" name="DOB"></label> <br /> <br />
<label><B>-Mobile No.: </B><input type="tel" name="Phone Number">
</label> <br /> <br />
<label><B> E-Mail: </B><input type="email" name="Name"></label> <br /><br />
<label for="Quota"><B> Quota: </B></label>
<select name="Quota" id="Quota:a">
<option value="Management;">Management</option>
<option value="Government;">Government</option>
</select> <br /> <br />
center><button type="submit" value="Submit">
< a href="first.html">Submit</a></button>
<button type="reset" value="Reset">Reset</button></center>
</fieldset>
</form>
</div>
</html>
```

Output

College Registration

Personal Details

Name:

Gender: M ☐ F ☐

Address:

DOB:

-Mobile No.:

E-Mail:

Quota: Management

Submit
Reset

Images: The HTML `<img>` tag is used to embed an image in a webpage.

* Images are not technically inserted into a webpage, images are linked to web pages.

* `<img>` tag is empty, it contains attributes only, and does not have a closing tag.

* `<img>` tag has two required attributes

src → specifies the path to the images

alt → specifies an alternate text for the image.

width → specifies the width of an image

height → specifies the height of an image

Syntax: `<img src = "html5.gif" alt = "HTML5 Icon" width = "128" height = "128">`

Image Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body style = "text-align: center;">
```

```
<h3> CITE Logo Image </h3>
```

```
<img src = " logo.png" width = "100"
```

```
height = "50" alt = " CITE logo image.png">
```

```
</body>
```

```
</html>
```

output:

CITE logo Image

logo..

Hyperlinks Example

Program

```
<!DOCTYPE html>

<html>
<head>
<title>CITCHENNAI</title>
</head>
<body>
<h1 face="purple" align="center"><font color="blue">
Welcome to Chennai CIT - The Best Engineering College in Chennai</font></h1>
<ul type="square">
<li><h2>Autonomous</h2></li>
<li><h3>NBA Accredited</h3></li>
<li><h4>Nirf 100th-150th ranking</h4></li>
<li><h5>NAAC A+</h5></li>
CIT has been approved by the AICTE, New Delhi, affiliated to Anna University
Chennai.</p>
<a href="https://www.citchennai.edu.in/training-placement/">CIT training and
placement</a><br><br><br>
<a href="mailto:info@citchennai.net">Reach us through e-mail</a>
</body>
</html>
```

OUTPUT

Welcome to Chennai CIT - The Best Engineering College in Chennai

- Autonomous
- NBA Accredited
- Nirf 100th^{-150th} ranking
- NAAC A+

CIT has been approved by the AICTE, New Delhi, affiliated to Anna University Chennai

[CIT training and placement](https://www.citchennai.edu.in/training-placement/)

[Reach us through e-mail](mailto:info@citchennai.net)

HTML Attributes: HTML elements have attributes that will provide additional information about the particular element.

* It takes 2 parameters ie [name and value] which define the properties of the element and are placed inside the element tag.

Points to remember for attributes:

* Attributes always come in name/value pairs like this attribute-name = "value"

* Attribute name and value are case sensitive and it should be written in lowercase only

* Attributes are always added to the start tag of an HTML element.

* Attribute value should be enclosed in quotes. Double quotes are most common, but single quotes are also allowed.

* In some rare case, when attribute value itself contains quotes, it is necessary to use single quotes. name = 'John "Shotgun" Nelson'

Syntax:

<element attribute-name = "value">content
</element>

Attributes types:

1) Id attribute: This attribute is used to provide unique identification to an element

<p id = "first paragraph">
This is first paragraph of text

</p>

2) class attribute: This attribute is used to indicate the class or classes that a tag might belong to that class.

* Like id, class is used to associate a tag with a name.

```
<p id = "first paragraph" class = "important">  
</p>
```

* first paragraph also indicate that this paragraph belongs to class group called important.

* The main use of class attribute is to relate a group of element to various stylesheet rules.

3). Style Attribute: This attribute is used to provide various CSS effects to the HTML element such as increasing font-size, changing font-family, coloring etc.

Syntax: <p style = "colored"> This is red.
paragraph </p>

4) Title Attribute: It is used to explain an element on hovering the mouse over it.
* The title attribute is used as text tooltip in most of browsers.

* It display its text when user move the cursor over a link or any text.

Syntax: <h1 title = "This is heading tag">

Example of title Attribute

```
</h1>
```

5). href attribute: This attribute is used to specify a link to any address.

* This attribute is used along with <a> tag
* This attribute provides hyperlink and if it is blank, then it will remain in same

Page.

Syntax:
This is a link

6). src attribute: It is one of the important and required attribute of element.
* This attribute can contain image in same directory or another directory.
* The image name or source should be same or else browser will not display the image.

Syntax:

7). alt attribute: It is used show or display something if the primary attribute (e) tag fails to display the value assigned to it.

Syntax:

8). width and height attribute: This attribute is used to adjust height and width of image.

Syntax:

9). lang attribute: This is used to declaring a language and it is important for accessibility applications and search engines.

Syntax: <html lang="en">
English Language </html>

Structuring page: In addition to defining individual part of your page (such as Paragraph or image) HTML also has a number of blocks level element used to define areas of your website (such as header, the navigation menu, the main

Content column).

Header: `<header>` It is usually a big strip across the top with a big heading, logo and perhaps a tagline.

Navigation bar `<nav>` Links to the sites main Section, usually represented by menu buttons, links or tabs. Like header this content usually remains consistent from one webpage to another having inconsistent navigation on your website will just lead to confused, frustrated users.

Main content: `<main>` with various content Subsections represented by `<article>` `<section>` `<div>` A big area in the center that contain most of the unique content of a given webpage. Eg:- Video you want to watch or the main story your reading, or the map you want to view, or the news headlines.

* This is the one part of websites that definitely will vary from page to page. Sidebar `<aside>`: often placed inside `<main>` some peripheral info, links, quotes, ads etc.. usually this is contextual to what is contained in the main content. Eg: on a news article page, the sidebar might contain the authors bio, or links to related article.

* But there are also cases where will find some recurring element like a secondary navigation system.

footer <footer>: A strip across the bottom of the page that generally contains line Point, Copyright notices, or contact info its a place to put common information (like the header) but usually, that information is not critical or secondary to the website itself.

* The footer is also sometimes used for SEO (search Engine optimization) is the process of making a website more visible in search results, also termed improving search ranking by providing links for quick access to popular content.

Program**Header**

```
<!DOCTYPE html>
<html>
<body>

<article>
  <header>
    <h1>A heading here</h1>
    <p>CIT</p>
    <p>Some additional information here</p>
  </header>
  <p>Chennai</p>
</article>

</body>
</html>
```

Output

A heading here

CIT

Some additional information here

Chennai

Program**NAVIGATOR**

```
<!DOCTYPE html>
<html>
<body>

<h1>The nav element</h1>

<p>The nav element defines a set of navigation links:</p>

<nav>
  <a href="/html/">HTML</a> |
  <a href="/css/">CSS</a> |
  <a href="/js/">JavaScript</a> |
  <a href="/python/">Python</a>
</nav>

</body>
</html>
```

Output**The nav element**

The nav element defines a set of navigation links:

[HTML](#) | [CSS](#) | [JavaScript](#) | [Python](#)

SECTION EXAMPLE

Program

```
<!DOCTYPE html>
<html>
<body>

<h1>The section element</h1>

<section>
  <h2>CIT</h2>
  <p>Chennai Institute of Technology is an industry-connected
autonomous institute, and is a co-educational engineering college
located at Kundrathur, Chennai, Tamil Nadu, India. It was established
in 2010 by the Parthasarathy Seeniammal Educational Trust.</p>
</section>

<section>
  <h2>Address: </h2>
  <p>Sarathy Nagar, Kundrathur, Chennai, Malayambakkam, Tamil Nadu
600069</p>
</section>

</body>
</html>
```

The section element

CIT

Chennai Institute of Technology is an industry-connected autonomous institute, and is a co-educational engineering college located at Kundrathur, Chennai, Tamil Nadu, India. It was established in 2010 by the Parthasarathy Seeniammal Educational Trust.

Address:

Sarathy Nagar, Kundrathur, Chennai, Malayambakkam, Tamil Nadu 600069

ASIDE EXAMPLE

Program

```
<!DOCTYPE html>
<html>
<body>

<p>Chennai Institute of technology, the best engineering
college in Chennai was established with the objective of
providing quality technical education with adequate
industrial exposure than any other engineering colleges
in Chennai, caters the needs of the youth with its
innovative teaching methods.</p>

<aside>
  <h4>Our Infrastructure and Facilities</h4>
  <p>For students who would like to study at an
engineering college with an established reputation for
high quality teaching and research, Chennai Institute of
Technology has a lot to offer.</p>
</aside>

</body>
</html>
```

OUTPUT

Chennai Institute of technology, the best engineering college in Chennai was established with the objective of providing quality technical education with adequate industrial exposure than any other engineering colleges in Chennai, caters the needs of the youth with its innovative teaching methods.

Our Infrastructure and Facilities

For students who would like to study at an engineering college with an established reputation for high quality teaching and research. Chennai Institute of Technology has a lot to offer..

Footer Example

Program

```
<!DOCTYPE html>
<html>
<body>

<h1>The footer element</h1>

<footer>
  <p>Author: CIT<br>
  <a
href="mailto:chennai.cit@gmail.com">chennai.cit@
gmail.com</a></p>
</footer>

</body>
</html>
```

Output

The footer element

Author: CIT
chennai.cit@gmail.com

Conclusion:

By following this basic html structure, web developers can create consistent, and well organized HTML document that are easy to understand and maintain.

HTML Semantic Elements:

* Semantic Element = Element with meaning.

* Semantic element defines its meaning accurately to both browser and developer.

Non-Semantic Element:

<div> Tells nothing about its content

Semantic Element:

<form> <table> <article> clearly defines its content.

Why to use semantic Element: It increases the accessibility of the website and provide better website structure.

* It allows data to be shared and reused across applications, enterprises and communities

Semantic Elements in HTML

`<article>` Element Specifies independent, self contained content eg. Magazine, newspaper, Blogspot etc.

`<aside>` Defines the content which is indirectly provide information to main content. It is frequently represent as sidebar.

`<details>` Defines additional information that the user can view or hide.

`<figcaption>` Defines a caption for `<figure>` element. It can be placed first or last child of `<figure>` element.

`<figure>`
Defines self contained content, like illustration, diagrams, photos, etc....

`<footer>`
Defines footer for a document or section.

Eg: Author info, copyright etc....

`<header>`
It is a container for introductory content or set of navigational link.

Eg: Logo, icon etc.

`<main>`
Specifies the main content of document.

`<mark>`
Defines marked/highlighted text.

`<nav>`
Defines a set of navigation link.

<section>

Defines a section in a document. It stores different element together.

Eg: chapter, Introduction, News item etc.

<summary>

Define a visible heading for <details> elements.

<time>

Defines a date/time.

Semantic Elements.

